



The burden of air pollution (PM_{2.5}) in subnational Australian life expectancy

Pattheera Somboonsin^{a,b,e,*}, Juthasinee Thanyapranedkul^c, Ivan C. Hanigan^{d,e,f}, Vladimir Canudas-Romo^b

^a Healthy Environment and Lives (HEAL) Global Research Centre, Health Research Institute, University of Canberra, Bruce, ACT, 2617, Australia

^b School of Demography, The Australian National University, Canberra, 2601, Australia

^c Department of Environmental Science, Faculty of Science and Technology, Thammasat University, Pathumthani Province, 12120, Thailand

^d WHO Collaborating Centre for Climate Change and Health Impact Assessment, Curtin School of Population Health, Curtin University, Western Australia, 6102, Australia

^e Healthy Environment and Lives (HEAL) National Research Network, Australia

^f Centre for Safe Air, NHMRC CRE, Australia

ARTICLE INFO

Keywords:

Australia
Life expectancy
Air pollution
Life-years lost
Subregion
Mortality
AirQ+

ABSTRACT

Australia is renowned for having some of the cleanest air globally, yet air pollution remains a persistent issue that significantly impacts life expectancy. This study quantifies the burden of annual ambient particulate matter (PM_{2.5}) in Australia's life expectancy: measuring life-years lost (LYL) from 2001 to 2019, considering geographical and demographic differences. Data were obtained from the Australian Bureau of Statistics, the Australian Human Mortality Database, and the Centre for Safe Air. Findings indicate that dust storms and bushfires were the main sources of PM_{2.5} during the 19-year period, with concentrations ranging from 3.39 to 13.78 $\mu\text{g}/\text{m}^3$. The Northern Territory showed the highest levels (6.38–13.78 $\mu\text{g}/\text{m}^3$), consistently exceeding both the national and WHO guidelines, while Tasmania had the lowest (4.07–5.73 $\mu\text{g}/\text{m}^3$), generally remaining below these thresholds. Males experienced higher LYL than females across all regions, with the largest gap observed in the Northern Territory in 2013–2.63 (95 % CI: 2.40–2.88) years for males versus 2.15 (95 % CI: 1.98–2.33) years for females. This region also exhibited the highest LYL in the country, while Tasmania had the lowest. Regional differences were observed, with rural areas in Queensland, South Australia, Western Australia, and Tasmania showing higher LYL than urban areas. In contrast, Greater Sydney had higher LYL than rural New South Wales. LYL was also greater among older individuals than adults. These results highlight the need for targeted interventions to address air pollution and its health impacts across different regions and demographics in Australia.

1. Introduction

Air pollution is a significant concern for populations worldwide. The Sustainable Development Goals (SDGs) emphasise this issue, with specific targets aimed at reducing mortality from environmental factors (Target 3.9), minimising the environmental impacts in urban areas (Target 11.6), and managing pollution and waste to decrease the release of pollutants that affect the global environment and human health (Target 12.4) (United Nations, 2015, 2024). Outdoor air pollution is a major contributor to the global burden of disease and a key factor in the decline of life expectancy worldwide (Cao and Wu, 2025; Lelieveld et al., 2020; Somboonsin et al., 2023). Specifically, particulate matter

less than 2.5 μm (PM_{2.5}), which arises from both man-made activities such as industry, transport, and agricultural burning, as well as natural events like dust storms and wildfires (Butler and Whelan, 2018; Hanigan et al., 2021; Pinichka et al., 2017), often exceeds the World Health Organization's (WHO) standard limits in several countries (Baklanov et al., 2016).

Despite Australia being ranked among the top ten countries for good air quality (IQAir, 2023; Lim et al., 2020), the country experiences frequent bushfires during the summer months and dust storms due to its dry landscape, leading to elevated levels of PM_{2.5} (Cao et al., 2023; Graham et al., 2021; Hill et al., 2021; Ryan et al., 2021). The impact of air pollution on the Australian population has received comparatively

* Corresponding author. School of Demography RSSS Building, 146 Ellery Crescent, The Australian National University, Acton ACT, 2601, Australia.

E-mail addresses: paire.somboonsin@anu.edu.au, paire.somboonsin@canberra.edu.au (P. Somboonsin).

<https://doi.org/10.1016/j.envres.2025.122690>

Received 5 May 2025; Received in revised form 15 August 2025; Accepted 23 August 2025

Available online 24 August 2025

0013-9351/© 2025 The Authors. Published by Elsevier Inc. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

less attention than in other regions, such as Africa and Asia. Existing studies on the long-term effects of PM_{2.5} on life expectancy are limited, especially at the subnational level, even though no Australian cities have consistently achieved the National Environment Protection Measures (NEPM) standard for PM_{2.5} (de Jesus et al., 2020; Hertzog et al., 2024). Additionally, there is a gap in Australian research focusing on how PM_{2.5} affects different genders and age groups, with limited studies exploring these variations comprehensively. Existing research has typically focused on specific genders, geographic areas, age groups, and causes of death (Chen et al., 2016; Dirgawati et al., 2019; Walter et al., 2021; Yu et al., 2020). However, studies comparing the overall impact of PM_{2.5} on all-cause mortality across subnational regions, and across different age and gender groups, are relatively scarce.

PM_{2.5} data can be accessed from several sources. Although ground station data is widely recognised as the most accurate and effective means of measurement (Palmer et al., 2018), it has limitations, particularly its inability to monitor all areas (Li et al., 2021; Zeydan and Wang, 2019). In recent years, the use of remote sensing techniques to assess PM_{2.5} levels via satellite has grown, owing to their cost-effectiveness and capability to monitor regions beyond the reach of ground-based stations (Buchard et al., 2016; Chen et al., 2021; Chu et al., 2016). NASA satellites, for example, allow researchers to average PM_{2.5} concentrations in specific areas of interest, such as national, sub-national, and sub-district levels (Chidburee et al., 2023). However, relying solely on ground station or satellite data may not fully capture the complexities of a large and diverse continent like Australia, which experiences annual bushfire events. In recent years, Australian researchers have developed methodologies to estimate daily PM_{2.5} levels from bushfire smoke using random forest modelling, which has proven to be particularly effective and relevant for the country's unique conditions (Borchers-Arriagada et al., 2024).

Although the WHO developed the Air Quality Health Impact Assessment Tool (AirQ+) to quantify mortality attributable to PM_{2.5} and support global research, no study has yet applied this tool to examine PM_{2.5} impact on Australia's life expectancy. AirQ+ software is highly effective for assessing both short- and long-term exposure to pollutants, including particulate matter, and has been widely used in many countries, particularly developed nations (De Marco et al., 2018; Ntourou et al., 2023; Ntourou et al., 2023; Rovira et al., 2020; Sicard et al., 2019). Most research in Australia, however, relies on epidemiological and statistical methods such as relative risk (Aragnou et al., 2021; Hanigan et al., 2021) and regression (Jegasothy et al., 2023; Yu et al., 2020), without utilising dedicated software like AirQ+. Given its widespread use in other developed countries, applying AirQ+ in Australia could provide valuable insights and allow for comparison between traditional statistical calculations and software-based estimates, offering a more robust understanding of the impacts of PM_{2.5}.

Previous research on health and mortality related to air pollution has often employed epidemiological measures such as years of life lost, disability-adjusted life years, and years lived with disability, or reported outcomes as attributed deaths (Aragnou et al., 2021; Bai et al., 2020; Broome et al., 2019). However, life-years lost (LYL), calculated using standard demographic techniques, offers a valuable approach for comprehensively understanding the number of years lost due to PM_{2.5}. LYL is the complement of life expectancy, and as the latter, it is measured in years. LYL avoids the confounding effects of population structure by using life table techniques (Andersen, 2013; Andersen et al., 2013; Houle et al., 2024; Somboonsin et al., 2023). Therefore, this study uses LYL to investigate the effects of PM_{2.5} in Australia.

The study aim is to analyse deaths attributable to PM_{2.5} across Australia, quantifying its impact on life expectancy from 2001 to 2019. This is achieved by employing AirQ+ software and recent PM_{2.5} data from the Clean Air and health Research Data and Analysis Technology (CARDAT) platform of the Centre for Safe Air. The effects of PM_{2.5} on mortality are examined by differentiating outcomes by sex, age, and geographic area, specifically within Greater Capital City Statistical

Areas (GCCSAs). This comprehensive approach aims to highlight regional disparities and temporal trends, guiding targeted public health interventions and informing policy formulation.

2. Methodology

2.1. Data

According to the Australian Bureau of Statistics (ABS) geographic classification, this study focuses on eight states and territories: New South Wales (NSW), Victoria (VIC), Queensland (QLD), South Australia (SA), Western Australia (WA), Tasmania (TAS), the Northern Territory (NT), and the Australian Capital Territory (ACT). For GCCSAs, areas were divided into greater city and rest of state for each state and territory, except for the ACT, which is not subdivided. More detailed information about Australian geography is provided in the supplementary materials (Appendix 1) and elsewhere (Australian Bureau of Statistics, 2021b).

This study utilised three primary data sources: the ABS, the Australian Human Mortality Database (AHMD), and the Australian Centre for Safe Air's data platform CARDAT. Together, these sources provided demographic information (age- and sex-specific deaths, population sizes, and mortality rates) as well as data on PM_{2.5} from 2001 to 2019. Population data for GCCSAs were obtained from the ABS (Australian Bureau of Statistics, 2022, 2023), which offers Australian population statistics. Population data refers to the Estimated Resident Population (ERP), the official estimate of the Australian population, which links individuals to their place of usual residence within the country. This data provides population sizes in five-year age groups up to age 85 and over, by sex in 2001–2019. Death counts by age (30–85 years) and sex, sourced from the Australian Institute of Health and Welfare (AIHW), were accessed through the Clean Air and health Research Data and Analysis Technology (CARDAT) platform (<http://cardat.github.io/>). CARDAT is an advanced IT infrastructure supported by the Centre for Safe Air and the World Health Organisation Collaborating Centre for Climate Change and Health Impact Assessment. In addition to population and health datasets, CARDAT provides data on air pollution in Australia. Using CARDAT, this study obtained particulate matter data from 2001 to 2019 by GCCSA. Daily PM_{2.5} estimates were generated via random forest modelling at a 5 × 5 km spatial resolution across the country. Further details on the PM_{2.5} modelling methodology can be found in the supplementary materials (Appendix 2) as well as in earlier publications (Borchers-Arriagada et al., 2024).

Age- and sex-specific death and population counts for all Australian states and territories from 2001 to 2019 were also obtained from the AHMD (AHMD, 2025) to refine estimates of five-year age-group mortality at the GCCSA level. This source includes information on deaths, population estimates, death rates, and life tables, categorised by single years and five-year age groups up to age 109, with an open age interval for those aged 110 and above. Mortality, population and life table data have been compiled by the AHMD using official ABS statistics from annual reports, following the methodological protocol of the international Human Mortality Database (AHMD, 2025; HMD, 2025).

Finally, the history of major bushfires and dust storms in Australia, used to indicate years with significant events in Fig. 1, was compiled from several sources (Aragnou et al., 2021; Bureau of Meteorology, 2019; Country Fire Authority, 2024; Forest Fire Management Victoria, 2023; Hanigan et al., 2018; NASA, 2006, 2009; NSW Office of Environment and Heritage, 2022; NSW rural fire service, 2014; Romsey Australia; The South Australian Country Fire Service, 2023). Details of all the data used are included in Table A2 in the supplementary information.

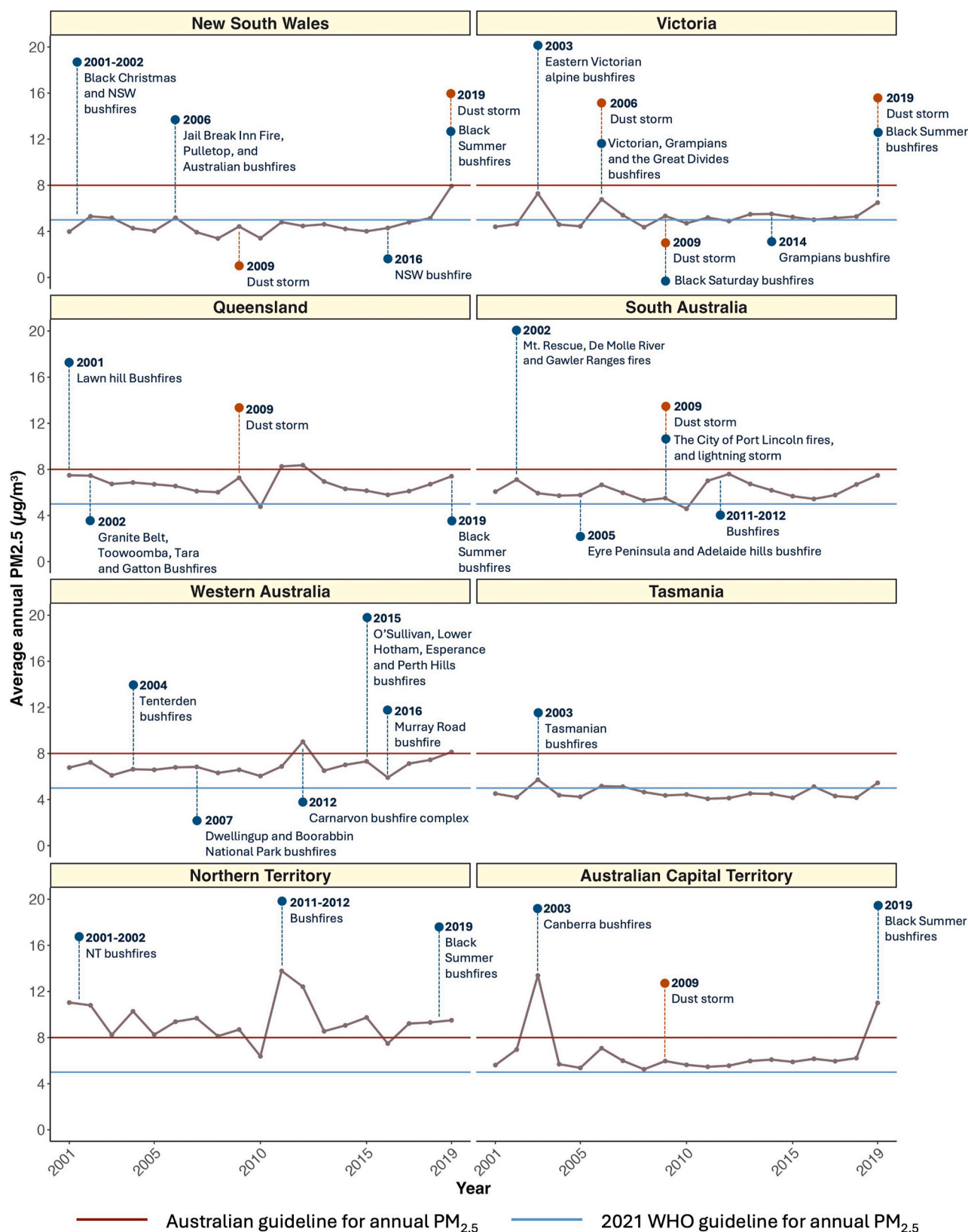


Fig. 1. Concentration of $PM_{2.5}$ and the main source of $PM_{2.5}$ in Australian states and territories from 2001 to 2019.

Note: The red lines represent the National Environment Protection for annual average $PM_{2.5}$ concentrations in Australia, set at $8 \mu g/m^3$. The blue lines represent the World Health Organisation (WHO) Global air quality guidelines for annual average $PM_{2.5}$ concentrations, updated in 2021, set at $5 \mu g/m^3$. Red dots indicate dust storms as the main source of $PM_{2.5}$, and blue dots for bushfires as the source. States and territories are ordered according to their numerical state codes defined by the Australian Bureau of Statistics (ABS), and this ordering is applied consistently throughout all figures (see code and state name in Table A1 in the Supplementary Information).

2.2. Methods

2.2.1. Deaths attributable to PM_{2.5}

The estimation of deaths attributable to PM_{2.5} was conducted using age-specific deaths or incidence rates, population sizes for those aged 30 and over, and average annual pollutant concentration levels. AirQ+ version 2.2, a software tool from the WHO designed for health risk assessment of air pollution, was used to estimate the number of deaths due to PM_{2.5}. The software is based on methodologies and exposure-response functions derived from epidemiological research and is capable of calculating both the number and proportion of health outcomes related to short- and long-term exposure. AirQ+ is widely used in public health research and policy analysis, as it allows researchers to quantify disease burden using local population data, mortality rates, and pollution levels. More details on the software and its methodology are provided in the supplementary information and elsewhere (Amini et al., 2024; World Health Organization, 2020a).

For demographic data, age- and sex-specific deaths and population sizes between 2001 and 2019 for each GCCSA were used as part of the input for calculating deaths attributable to PM_{2.5}. ABS population data in five-year age groups and an open age interval for those aged 85 and above, were used after disaggregating the information up to age 100. Death counts by broad age groups were available from AIHW. However, to align with the population data and enable a detailed age-specific analysis across population and mortality datasets, we employed the AHMD, which provides death counts by states in five-year age groups up to 110+ years. The proportions of deaths from AHMD were then applied to estimate the number of deaths in five-year age groups from ages 30–100 years by age and sex for each GCCSA. Further details of the approach followed for disaggregating the information into each GCCSAs can be found in the supplementary materials (Appendix 3).

To obtain the input PM_{2.5} concentration levels, the Raster package in R statistical software was used. Files downloaded from the Bushfire-Specific PM_{2.5} in Australia Version 1.3 dataset, available on the CAR-DAT platform (Hanigan et al., 2023), were overlaid with ABS boundaries from the Australian Statistical Geography Standard (ASGS) Edition 3 (Australian Bureau of Statistics, 2021a) to ensure accurate geographic matching. The average PM_{2.5} concentration levels were then computed for both state and territory and GCCSAs levels. While some studies use population-weighted PM_{2.5} to estimate exposure levels (Broome et al., 2020; Hertzog et al., 2024)—adjusting for where people live to better reflect actual population exposure—AirQ+ relies on area-level annual mean concentrations to estimate deaths attributable to PM_{2.5}. When multiple geographic areas are entered, the tool applies population weighting across these areas to estimate the overall health burden (World Health Organization, 2020b). Our analysis used the standard approach recommended for AirQ+, using annual mean PM_{2.5} concentrations along with age- and sex-specific population and mortality data.

The deaths attributable to PM_{2.5} were calculated based on the number of deaths by age, relative risk of 1.08 (95 % CI: 1.06, 1.09) (Chen and Hoek, 2020) and a cutoff value of annual PM_{2.5} concentrations of 5 µg/m³, consistent with the WHO air quality guideline. This approach assumes negligible risk below this threshold, which may lead to an underestimation of health impacts at lower exposure levels. Recent cohort studies in Sydney such as (Hanigan et al., 2019) have found associations between long-term PM_{2.5} exposure and all-cause mortality even at lower concentrations (mean: 4.49 µg/m³; range: 2.78–13.81 µg/m³).

Our study introduces LYL as an alternative measure to assess the impact of PM_{2.5} on life expectancy. Since LYL is based on life expectancy, it incorporates demographic composition and does not require population weighting. By applying LYL, we provide a comprehensive perspective on the effects of pollution on life expectancy.

2.2.2. Life-years lost attributable to PM_{2.5}

LYL is the complement of life expectancy. In this study, LYL

attributable to PM_{2.5} measures the years lost associated with PM_{2.5} in the population between ages 30 and 100 years. We calculated LYL by using age-specific death rates for those ages of the study.

To start the LYL calculation, we first calculated life expectancy between ages 30 and 100, denoted as ${}_{70}e_{30}$, following life table notation, where ${}_ye_x$ represents the life expectancy between ages x and $x + y$. The input of the standard life table calculation is the age-specific mortality rates (Preston et al., 2000). Mortality rates for GCCSAs were calculated using the estimated age- and sex-specific death counts, divided by the ERP age- and sex-specific population in each GCCSA and time.

The total number of years between ages 30 and 100 (70 years) is equal to the sum of life expectancy, ${}_{70}e_{30}$, and LYL (70 = life expectancy + LYL) as shown in the following equation:

$$70 = {}_{70}e_{30}^{s,g,st,t} + {}_{70}LYL_{30}^{s,g,st,t}, \quad (1)$$

where ${}_{70}e_{30}^{s,g,st,t}$ represents the number of years lost for all causes from ages 30 to 100 by sex s , GCCSA g , state st , and time t . LYL in equation (1) can be further separated into LYL attributable to PM_{2.5} and to other risk factors. In the remainder of this study, we focus only on PM_{2.5}-attributable LYL. The LYL can also be broken down by age group to determine their respective age-specific allocations as follows:

$$70 = {}_{70}e_{30}^{s,g,st,t} + {}_{35}LYL_{30}^{s,g,st,t} + {}_{35}LYL_{65}^{s,g,st,t}. \quad (2)$$

At the state level, we combined LYL for greater cities and their corresponding rest of state regions within each state to obtain LYL estimates by sex and state. Detailed procedures for calculating LYL have been documented in previous studies (Andersen et al., 2013; Canudas-Romo et al., 2015; Erlangsen et al., 2017; Houle et al., 2024; Somboonsin et al., 2024). All analyses were conducted using R statistical software version 4.0.3. Further details on additional calculations are provided in the supplementary information.

3. Results

Between 2001 and 2019, air particulate matter (PM_{2.5}) in Australia primarily originated from dust storms and bushfires. Fig. 1 shows that the time trends of PM_{2.5} in Australia fluctuated over the 19-year period, with peak levels corresponding to natural events. For example, in 2009, Australian dust storms significantly affected New South Wales (NSW), Queensland (QLD), South Australia (SA), and Australian Capital Territory (ACT), causing a noticeable increase in PM_{2.5} levels compared to the previous year. Similarly, the major bushfires in 2019, known as the Black Summer bushfires, led to high PM_{2.5} levels in many states.

The national standard for annual average PM_{2.5} concentrations in Australia is set at 8 µg/m³. While most regions had annual average PM_{2.5} levels below this standard, the Northern Territory (NT: 6.38–13.78 µg/m³) frequently exceeded it in many years. The ACT also recorded levels above this standard in specific years with major bushfires, such as 2003 (13.39 µg/m³) and 2019 (11.01 µg/m³).

In comparison to the WHO guideline, which sets the annual average PM_{2.5} concentration at 5 µg/m³, many states generally stayed above this threshold level throughout the years. However, Tasmania (TAS) and NSW often reported annual PM_{2.5} levels below the WHO guideline, with occasional increases during major events. Even during these years, their levels remained relatively close to the WHO standard. For example, in 2003, PM_{2.5} concentrations reached 5.17 µg/m³ in NSW and 5.73 µg/m³ in TAS, while in 2019, these values rose to 7.93 µg/m³ in NSW and 5.45 µg/m³ in TAS. It is important to note that the WHO guideline is shown as an international benchmark to contextualise long-term exposure trends, especially in contrast to Australia's national standard of 8 µg/m³. However, the state-level averages in Fig. 1 can still be exceeded by some specific areas within states.

Fig. 2 shows that LYL due to PM_{2.5} trends across states and territories from 2001 to 2019. Males consistently had higher LYL than females in

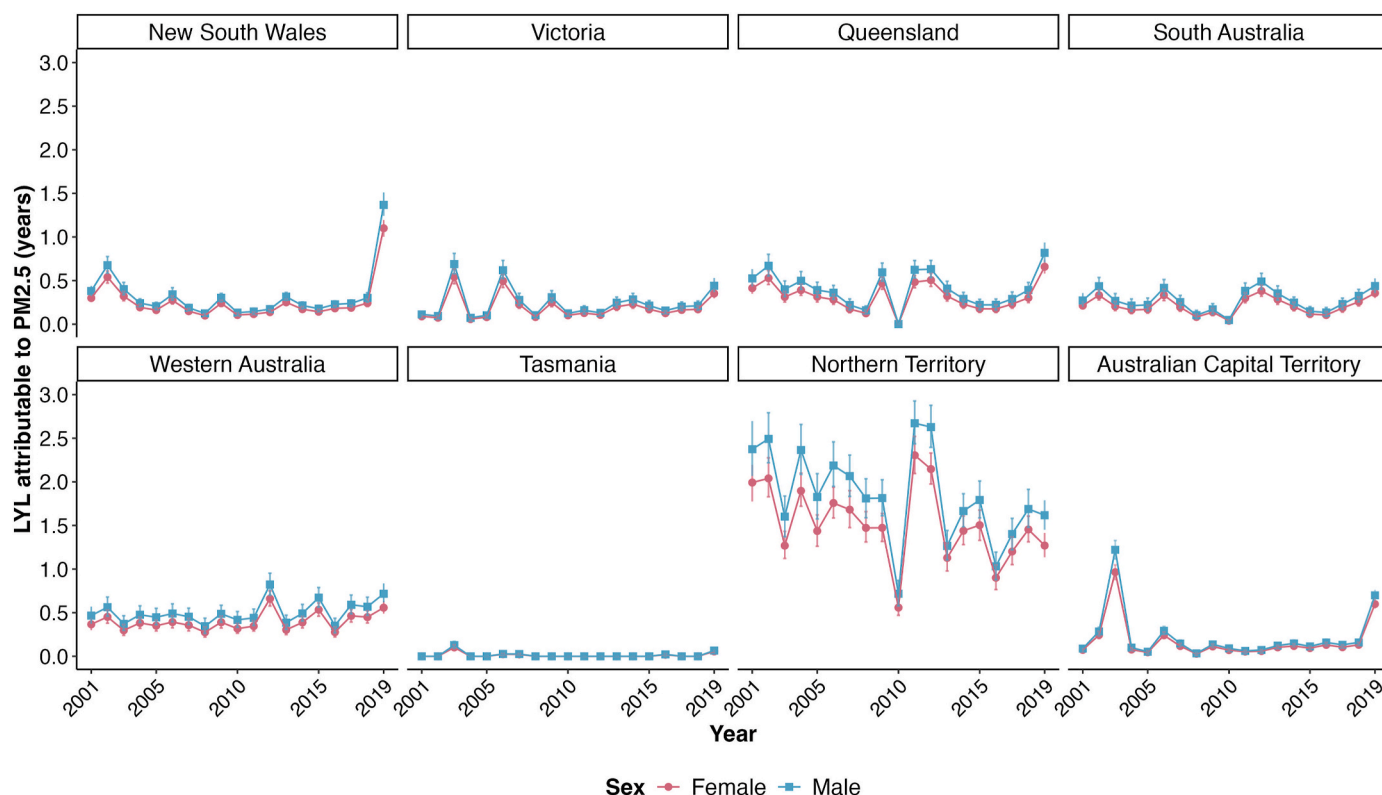


Fig. 2. Life-years lost attributable to $PM_{2.5}$ by sex, 2001–2019.

all regions. In the NT, although LYL fluctuated, there was a general decreasing trend from 1.99 (95 % CI: 1.78, 2.20) years in 2001 to 1.27 (95 % CI: 1.14, 1.41) years in 2019 for females, and from 2.37 (95 % CI: 2.07, 2.70) years in 2001 to 1.62 (95 % CI: 1.45, 1.79) years in 2019 for males. Despite this decrease, LYL in the NT remained higher compared to other states, with a noticeable gender gap reaching a maximum of 0.48 years in 2012.

In QLD, SA and Western Australia (WA), LYL fluctuated, with significant increases during major dust storms or bushfires, such as in 2002, 2006, 2009, 2012, 2015 and 2019. NSW, Victoria (VIC) and the ACT generally had LYL below 0.3 years in many years. However, higher levels were observed during major events, such as in 2003 in the ACT, with LYL reaching 0.97 (95 % CI: 0.88, 1.05) years for females and 1.22 (95 % CI: 1.12, 1.33) years for males, and 2019 in NSW, with LYL at 1.10 (95 % CI: 1.00, 1.19) years for females and 1.37 (95 % CI: 1.24, 1.51) years for males.

TAS showed relatively lower and more stable LYL compared to other states. Even during significant years, such as 2003 and 2019, LYL in TAS remained low, with values of 0.10 (95 % CI: 0.07, 0.13) years for females and 0.13 (95 % CI: 0.09, 0.17) years for males in 2003, and 0.06 (95 % CI: 0.04, 0.08) years for females and 0.07 (95 % CI: 0.04, 0.09) years for males in 2019.

In 2019, during the Black Summer bushfires, elevated levels of $PM_{2.5}$ were recorded across many regions of Australia, with particularly high concentrations in the ACT and NT (see Fig. A2 in the supplementary information for detailed $PM_{2.5}$ distribution by state). At the GCCSA level, as shown in Fig. 3, significant regional variation in $PM_{2.5}$ concentrations was observed. Urban areas such as Greater Sydney, Greater Darwin, Greater Brisbane, and the ACT experienced particularly high levels of $PM_{2.5}$. Additionally, rural regions in the NT and WA exhibited concentrations exceeding $8 \mu g/m^3$, highlighting the widespread impact of the bushfires on air quality across the country.

Fig. 4 presents LYL attributable to $PM_{2.5}$ by GCCSAs. Similar to Fig. 2, there was no clear trend from 2001 to 2019. LYL in greater cities was consistently higher than in regional areas of NSW every year, with a significant gap in 2019 when Greater Sydney recorded 0.87 (95 % CI: 0.81, 0.94) years of LYL compared to 0.36 (95 % CI: 0.32, 0.42) years in the rest of NSW. For most years, LYL in the rest of NSW was close to 0, except in 2019. In VIC, Greater Melbourne generally had higher LYL than the rest of the state, except in 2003 when LYL levels were practically the same across the state.

In contrast, LYL in the rest of the state was higher than in greater cities in QLD, SA, WA, and TAS. Notably, LYL in Greater Tasmania remained close to 0 in most years.

In the NT, LYL fluctuated with a general decrease over the study period. Greater Darwin had higher LYL than rural areas from 2003 to 2010. However, in 2011, LYL in rural areas surpassed that of Greater Darwin, and no significant differences between urban and rural areas were observed thereafter.

Finally, the ACT, which is a small geographic area without distinct urban and rural divisions, exhibited high LYL in 2003 and 2019 at 1.08 (95 % CI: 0.99, 1.19) and 0.65 (95 % CI: 0.59, 0.71) years, respectively, corresponding to major bushfires in these two years.

When examining age differences, adults aged 30–64 years had lower LYL compared to the older group aged 65 to 100 in all regions. In 2019, Fig. 5 indicates that the older group in Greater Sydney experienced the highest LYL from $PM_{2.5}$ pollution, accounting for 0.83 (95 % CI: 0.77, 0.89) years. The older group in Greater Darwin ranked second with 0.78 (95 % CI: 0.70, 0.86) years, followed by the ACT and rest of the NT, which had LYL values of 0.62 (95 % CI: 0.56, 0.68) and 0.58 (95 % CI: 0.50, 0.67) years, respectively. In contrast, no LYL was observed in Greater Hobart in 2019, while the LYL for older adults in the rest of TAS were at a similar level to the LYL for adults in both urban and rural areas of the NT.

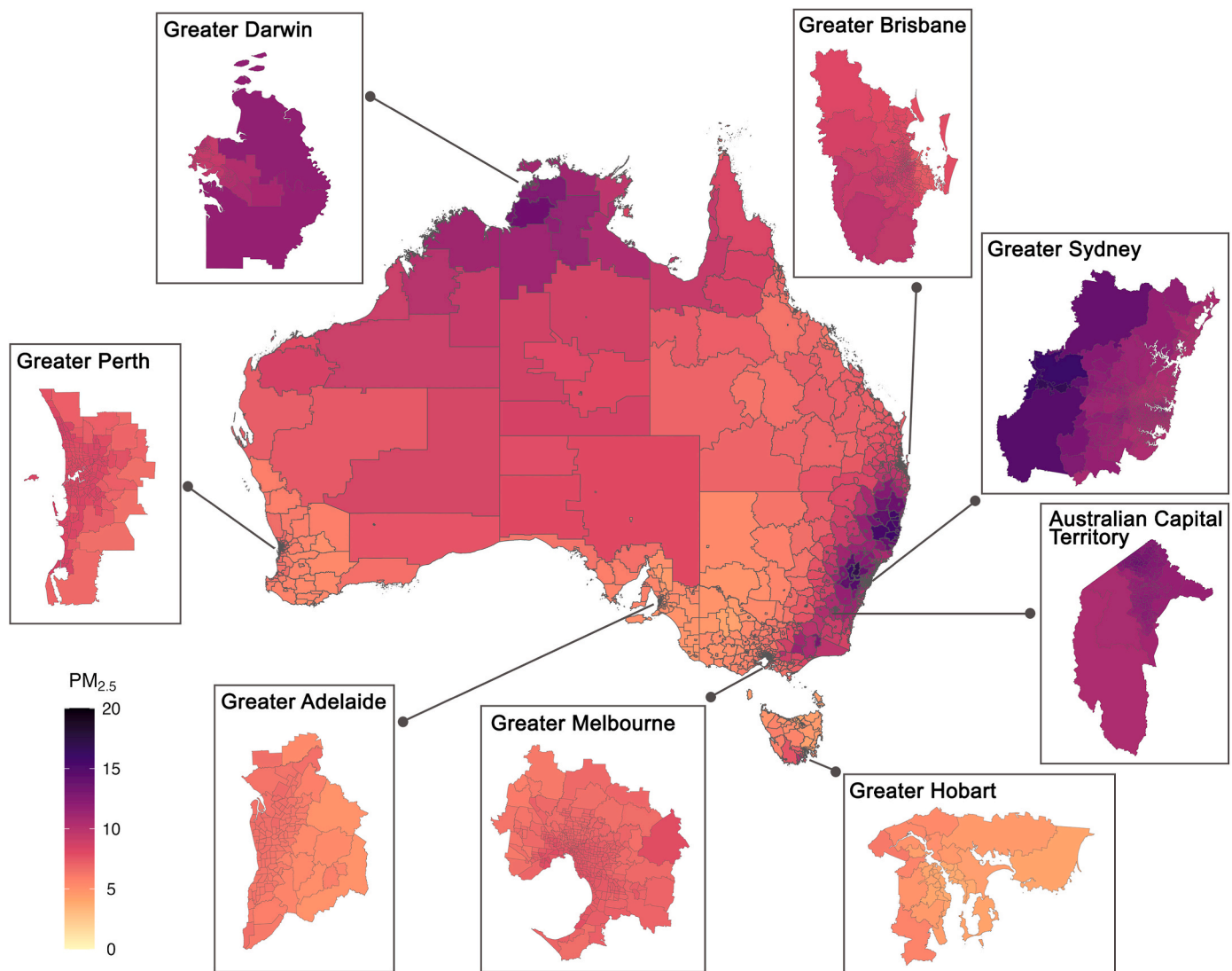


Fig. 3. Concentration of $PM_{2.5}$ in 2019 across Australia.

Note: $PM_{2.5}$ levels are measured in $\mu g/m^3$. Details on Australian states and territories are provided in the supplementary information.

4. Discussion

This study provides a novel contribution to Australian studies of pollution by examining the effects of $PM_{2.5}$ on life expectancy and life-years lost. By integrating a recent Australian $PM_{2.5}$ dataset with the AirQ+ model to estimate deaths attributed to $PM_{2.5}$, our approach provides an accessible method for calculating LYL. AirQ+ applies a log-linear risk model using relative risks derived from recent epidemiological evidence. We used the relative risk estimate from [Chen and Hoek \(2020\)](#), which provides a comprehensive meta-analysis of long-term $PM_{2.5}$ exposure and all-cause mortality based on high-quality global cohort studies ([Hanigan et al., 2021](#); [Egerstrom et al., 2023](#)), and is widely used in health impact assessments, including those by the WHO. The study is particularly distinctive in its focus on the Australian context, addressing regional disparities and subpopulations that are often underrepresented. Additionally, the integration of time trends allows for a clearer understanding of how the effects of $PM_{2.5}$ have evolved over time and provides a basis for identifying potential drivers behind these impacts. Previous studies by the Australian Burden of Disease Study (ABDS) and related national-level research have provided essential estimates of mortality and disease burden attributable to air pollution ([Hadei et al., 2020](#); [Lelieveld et al., 2020](#); [Hanigan et al., 2021](#);

[Hertzog et al., 2024](#)), using metrics such as Disability-Adjusted Life Years (DALYs), lost life expectancy at birth, and Years of Life Lost (YLL) to inform policy and interventions. Our study contributes to the existing literature by introducing an innovative methodological approach, namely LYL, the complement of life expectancy, to assess the effect of $PM_{2.5}$ on subnational Australian population. Similar to life expectancy, LYL is also measured in years, and it can be decomposed into its attributable to $PM_{2.5}$ and other risk factor components. This approach also translates complex epidemiological data into a format that is more comprehensive and accessible, not only for policymakers but also for the general public, thereby helping to raise awareness and strengthen community understanding of the health impacts of air pollution.

Our findings indicate that the LYL burden attributable to $PM_{2.5}$ had a maximum value of 2.30 years (95 % CI: 2.10–2.52) for females, and 2.67 years (95 % CI: 2.44–2.93) for males across Australian states and territories between 2001 and 2019. However, there was great heterogeneity across locations and years. This study enhances comprehension of the health impacts of air pollution and provides crucial insights for shaping public health interventions and policies. Using $PM_{2.5}$ estimates from CARDAT, our findings revealed that annual $PM_{2.5}$ levels across Australian states and territories ranged from 3.39 to 13.78 $\mu g/m^3$ between 2001 and 2019. While several states and territories exceeded the WHO's

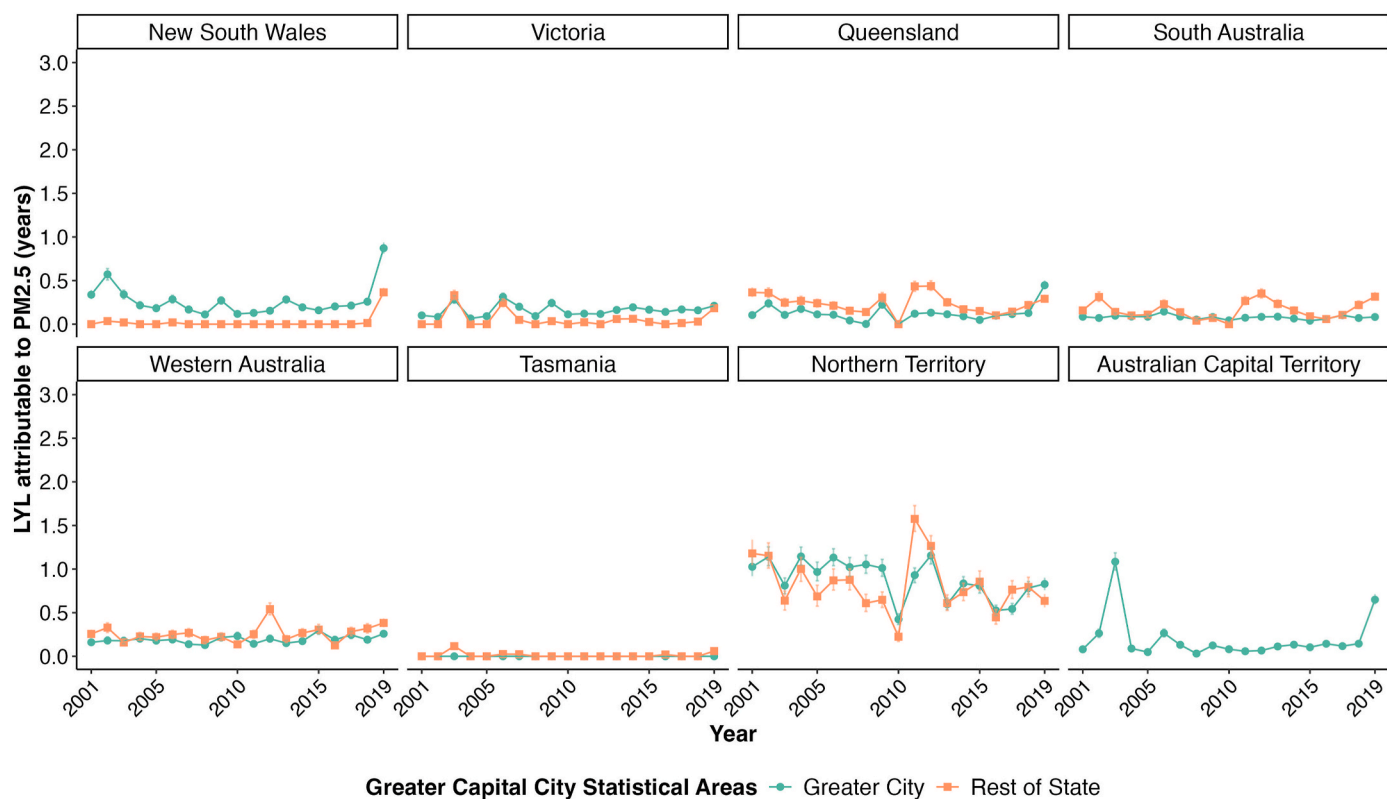


Fig. 4. Life-years lost attributable to $PM_{2.5}$ by Greater Capital City Statistical Areas (GCCSAs) and rest of state, Australia, 2001–2019.

Note: Names of Greater cities in each state and territory: the first row, from left to right – Greater Sydney, Greater Melbourne, Greater Brisbane, Greater Adelaide; the second row, from left to right – Greater Perth, Greater Hobart, Greater Darwin, Australian Capital Territory.

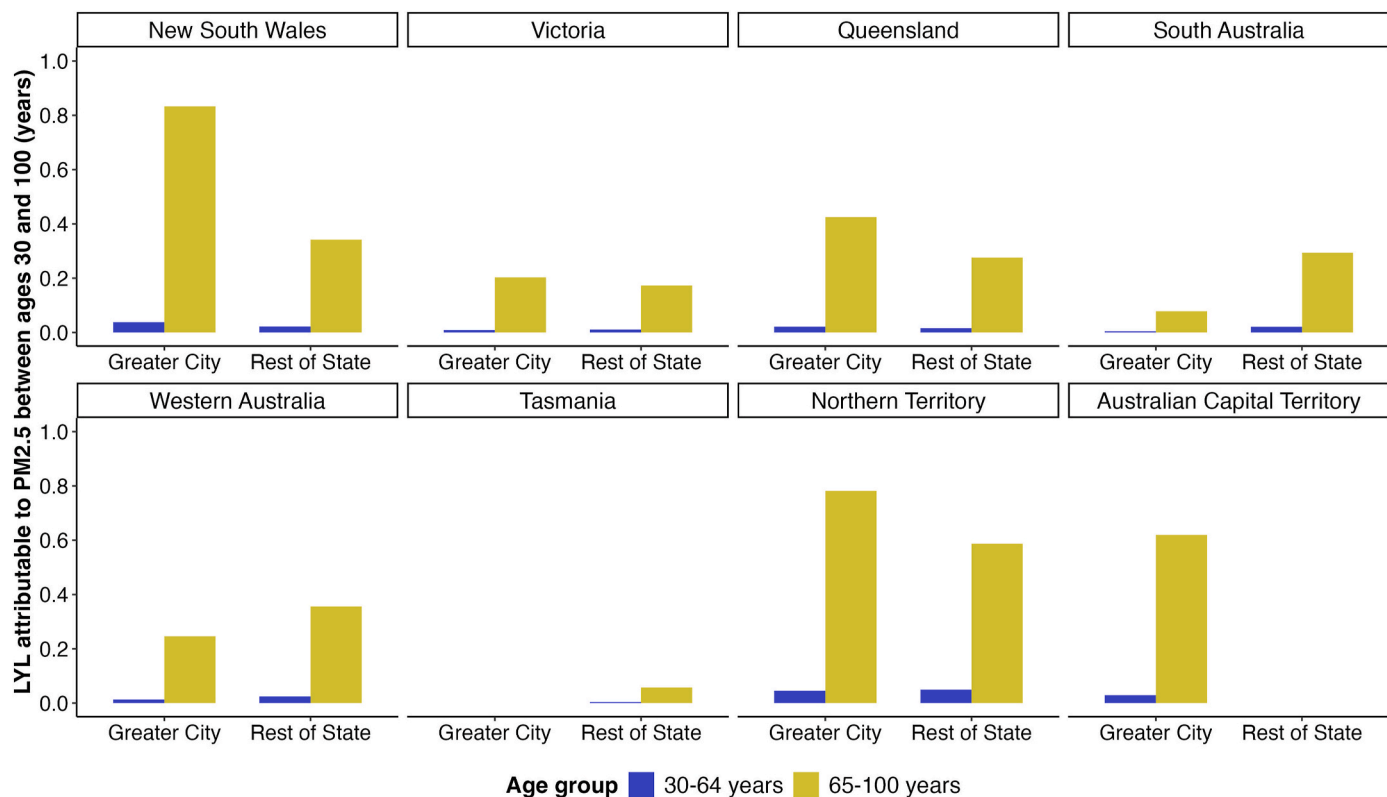


Fig. 5. Age-group life-years lost attributable to $PM_{2.5}$ by Greater Capital City Statistical Areas (GCCSAs) and rest of state, Australia, 2019.

Note: Names of Greater cities in each region: the first row, from left to right – Greater Sydney, Greater Melbourne, Greater Brisbane, Greater Adelaide; the second row, from left to right – Greater Perth, Greater Hobart, Greater Darwin, ACT.

2021 guideline for annual average PM_{2.5} concentrations (5 µg/m³) in multiple years, NSW and TAS generally remained below this threshold. In contrast, when compared to Australia's national air quality standard (8 µg/m³ annual average), only the NT exceeded this limit in 17 of the 19 years. In the NT, nearly all PM_{2.5} pollution in Darwin results from bushfires, which burn continually for six months each year. Although extreme fires may have been noted in certain years, these levels are generally consistent year after year due to regular early dry-season savanna burning, particularly in the western Top End (Bowman et al., 2024).

The relationship between elevated PM_{2.5} levels and major dust storms and bushfire events over the 19-year period studied was assessed. The results consistently showed that years and states with higher PM_{2.5} levels coincided with the occurrence of these events. For example, the ACT generally experienced stable and low PM_{2.5} concentrations throughout most years. However, significant bushfires, such as the Canberra bushfires in 2003 and the Black Summer bushfires in 2019 caused PM_{2.5} levels in the ACT to spike, surpassing those in other regions during these years. Additionally, the major dust storm event known as the Red Dawn in 2009 affected several states in eastern Australia, including NSW, VIC, QLD, SA, and the ACT, originating from the desert regions of Central Australia, leading to elevated PM_{2.5} levels across these regions (Aragnou et al., 2021).

Although SA, with its dry landscape, hot summers, and low average rainfall, frequently experiences dust storms and wildfires (Chen et al., 2016), PM_{2.5} levels estimated using smoke PM_{2.5} data from CARDAT were found to be below the Australian NEPM standard. However, when compared with satellite data, PM_{2.5} levels in South Australia were notably higher than those estimated by CARDAT and were among the highest in the country (see comparative PM_{2.5} data in Figs. A1 and A8 in the supplementary information). This demonstrates the differences between data sources and highlights the suitability of CARDAT for assessing PM_{2.5} levels within the unique environmental context of Australia.

Australia's dry climate and fire-prone landscape make it particularly vulnerable to both planned burns and unplanned bushfires, which are major episodic contributors to PM_{2.5} concentrations, especially during severe fire seasons. Although our study did not disaggregate PM_{2.5} exposure by source, the CARDAT dataset captures both chronic and event-related pollution, including bushfire smoke. Consequently, some of the elevated LYL observed may have been driven by bushfire episodes.

LYL in the NT was higher than in other regions, which corresponds with their elevated PM_{2.5} levels. There is a strong positive correlation between PM_{2.5} concentrations and LYL across states and territories, indicating that regions with higher exposure tend to experience heightened mortality burdens (see Supplementary Figs. A9 and A10). As mentioned earlier, NT has dry landscapes and desert areas, making it more susceptible to natural events that increase PM_{2.5} concentrations. The dry conditions, particularly during the hot summer months, can lead to increased wind-blown dust, raising particulate matter levels in the air. Additionally, the NT experiences frequent burning every half-year, which significantly raised PM_{2.5} concentrations. These elevated levels contribute to higher health risks and mortality rates from respiratory and cardiovascular diseases, as well as exacerbations of asthma caused by particulate matter (Bowman et al., 2024; Chen et al., 2016; Hansen et al., 2012; Johnston et al., 2021).

Sex disparities were evident in our study, with Australian males losing more life years due to PM_{2.5} exposure than females across all regions. These findings align with PM_{2.5}-related mortality trends observed in several other countries (Chen et al., 2022; Johnston et al., 2013). Generally, men have higher mortality rates which contribute to a lower life expectancy at birth and higher LYL compared to women. The effect of PM_{2.5} on those higher male mortality rates can be explained by several factors. Occupational exposure is an important factor, as men are more likely to work in outdoor jobs such as coal mining and spend more

time outside, increasing their exposure to PM_{2.5} (Johnson et al., 2019; Regan and Partridge, 2013). In addition, biological and behavioural differences, including higher smoking rates and lower engagement with health services, may further increase men's vulnerability to pollution-related LYL (Parslow et al., 2004; Liu et al., 2021b).

Unlike in many other countries, rural areas of Australia tend to experience worse air quality than urban areas (Hill et al., 2021). The recent major bushfires in 2019–2020 also severely impacted rural areas, including the outback in SA, Namadgi in the ACT and the Orbost region in VIC, significantly contributing to elevated PM_{2.5} levels during those periods. The occurrence and length of extreme fire weather events have risen across vast regions of Australia, particularly in the southern and eastern areas and national parks, with dry eucalypt forests (Bowman et al., 2021; Cheney et al., 2012; Collins et al., 2021; Godfree et al., 2021). This trend is largely driven by climate change, as rising temperatures, decreased rainfall, and lower atmospheric moisture levels lead to drier landscapes (Hill et al., 2021).

It is generally understood that the effects of PM_{2.5} on health outcomes are more pronounced in urban areas compared to rural ones (Liu et al., 2021a), and our study found that Greater Sydney had higher LYL-related PM_{2.5} than the rest of the state over the period studied. This is consistent with previous research that attributes higher PM_{2.5} concentrations in the Greater Metropolitan Region of Sydney due to emissions from transport, industrial processes, and shipping activities that rely on low-quality diesel fuel (Broome et al., 2019; Hanigan et al., 2021). Additionally, LYL in Greater Melbourne was also higher than in rural areas in Victoria in many years. A possible explanation is that wood heaters, vehicles, power stations, and planned burns contribute to over 50 % of PM_{2.5}-related mortality and morbidity in both Greater Sydney and Greater Melbourne (Bailie et al., 2023; Broome et al., 2020).

Although much of the PM_{2.5} pollution in cities is attributed to traffic (Hadei et al., 2020; Li et al., 2017), the reduction in private car use and change to electrical transportation alternatives in Australian cities, along with increased public transport usage and a shift toward working from home, possibly contributed to lower PM_{2.5} levels in urban areas (Healy, 2012). This may explain why some greater cities exhibited lower LYL due to PM_{2.5} compared to the rest of their state. In addition to wind patterns, particularly during summer dust storms in desert areas, several factors likely contributed to the higher PM_{2.5} levels in rural areas of South Australia, leading to higher LYL compared to urban areas. These include industrial and mining activities prevalent in rural regions, emissions from wood heaters, farming, and generally higher mortality for all causes in rural populations (Fielke and Bardsley, 2015; Government of the Commonwealth of Australia - Australian Bureau of Stats, 2023; Haswell and Bethmont, 2016; Knibbs et al., 2018). Another possible factor is related to the definition of GCCSAs by the ABS. For example, in Queensland, excluding densely populated areas like the Sunshine Coast and Gold Coast from the greater city definition might have skewed the results, as these towns have significant populations, tourist attractions, and transportation patterns similar to Brisbane (Baum et al., 2009; Cunningham et al., 2019; Golam Mortoja and Yigitcanlar, 2021; Hefferan and Mannix, 2011). Further details on the GCCSA definitions can be found in the Supplementary Information Table A1. LYL by gender in GCCSAs followed a similar pattern with the state level, with males consistently showing higher LYL than females across both cities and the rest of the state (see Figs. A6 and A7 in the supplementary information for details).

Comparing LYL due to PM_{2.5} among adults and the older group revealed that the older group experienced higher LYL, which is consistent with findings from numerous previous studies (Yu et al., 2020; Zhao et al., 2022). Given that Australia is an ageing society with one of the highest life expectancies in the world, the proportion of people aged 65 years and older is significant (22.1 % for females and 19.2 % for males in 2023) (Australian Bureau of Statistics, 2024). Therefore, the need to improve air quality to protect the health of older people is both crucial and urgent.

To mitigate the impact of PM_{2.5} and improve air quality in Australia, coordinated efforts are required at all levels, including individuals, communities, and governments across state and national levels. Effective management of natural events such as bushfires and dust storms is critical. Government strategies should prioritise the careful use of planned burns to manage fuel loads and reduce the overall harm from extreme fires and associated smoke, including pollution impacts on populated areas. While planned burns cannot prevent bushfires, they reduce fuel availability, making fires easier to control and less severe in impact (Hertzog et al., 2024). However, regular back-burning can also elevate PM_{2.5} levels and negatively impact public health. Therefore, striking a balance between reducing major fire events and minimising air pollution from planned burns is essential and should be guided by health impact assessments and modelling. Additionally, developing advanced early warning systems can alert communities to areas with elevated PM_{2.5} levels, helping to reduce exposure (Zhao et al., 2022). Public awareness campaigns are equally important, informing individuals about the health risks associated with PM_{2.5} and promoting protective measures. Furthermore, human activities that contribute to climate change and environmental degradation play a role in exacerbating air pollution and altering natural processes. Climate variability, such as the influence of El Niño and La Niña, can further influence air quality and environmental conditions. At the individual level, we must foster awareness and behavioural changes (Williamson et al., 2018). For instance, reducing practices like agricultural burning, minimising waste, and transitioning to cleaner energy sources can help protect the environment. Governments can further support these efforts by encouraging the use of renewable energy, promoting sustainable agricultural practices, enforcing stricter emissions standards for industries, and funding research on environmental health, particularly for studies focusing on vulnerable populations disproportionately affected by air pollution. Together, these actions can mitigate human-driven contributors to PM_{2.5} pollution, protecting both public health and the environment.

This study has several limitations that should be considered. First, even though many studies indicate that mortality and health effects from air pollution are significant for children, AirQ+ excludes individuals under 30 years of age from its estimations, which lead to an underestimation of the impact of PM_{2.5} on populations. The long-term health effects of PM_{2.5} exposure on individuals, typically will accumulate over time and will contribute to the observed deaths at older ages. This is also consistent with modelling tools such as AirQ+, which limit long-term impact assessments to those aged 30 and above due to a stronger evidence base for these age groups. Secondly, population data for GCCSAs from the ABS are only available in five-year age groups up to age 85. As Australia is an ageing society, this restricting availability of data does not allow us to fully capture the impact of PM_{2.5} on older populations beyond the age of 85. We mitigated some of this shortcoming by disaggregating the population data up to age 100, though actual data would provide greater precision. Third, public data for age- and sex-specific deaths in smaller age intervals, such as five-year age groups, as well as life tables for GCCSAs, were not available. Although we verified life expectancy trends using data produced by the ABS, which showed similar patterns (see Fig. A3 in the supplementary information), the absence of detailed data may limit the precision of our findings at the GCCSA level. Also, this study used the same relative risk for all sex and age groups when estimating deaths attributable to PM_{2.5}. Future research could explore incorporating stratified relative risks, provided these are derived from epidemiological studies with strong validity for Australian populations to better capture biological and social differences in vulnerability. Moreover, future studies using finer spatial units such as Statistical Area Level 3 (SA3) would allow for a more detailed assessment of intra-state variation in both air pollution exposure and associated health outcomes. Such granularity would enhance the precision of burden estimates and support more targeted public health interventions at the local level. However, this detailed analysis would need to be accompanied by uncertainty calculations, given the

sensitivity of life expectancy and LYL to population size. Additionally, while this study presents trends in annual PM_{2.5} and LYL from 2001 to 2019 and shows a strong correlation between PM_{2.5} and LYL (Supplementary Figs. A9 and A10), we acknowledge that a time-series analysis could offer deeper insights into temporal patterns and drivers of change. Future research may build on our findings by applying time-series modelling techniques to explore the drivers of year-to-year variation in exposure and health burden, particularly in the context of policy changes, environmental events, and demographic shifts. Finally, to understand the impact of PM_{2.5} from bushfires and dust storms in Australia, further research is needed to incorporate temporal and seasonal factors, wind direction, and monthly analyses of PM_{2.5}. This approach would help capture the duration and intensity of bushfires and dust storms more accurately. Alternatively, focusing on PM_{2.5} levels specifically from these sources could better reflect their impact on the population.

5. Conclusion

In summary, this study provides a comprehensive assessment of how PM_{2.5} pollution has affected Australian life expectancy over 19 years. Bushfires and dust storms are closely linked to elevated levels of PM_{2.5} in the country, with years experiencing these events showing increase LYL attributable to PM_{2.5}. Our findings indicate that LYL among men were higher than among women across all regions. The NT exhibited the highest LYL, with no significant difference between urban and rural areas, while TAS had the lowest LYL overall. Greater Sydney had higher LYL compared to its rural areas. Although the ACT had generally low LYL, it experienced some of the highest levels in the nation during 2003 and 2019 due to major bushfires. Furthermore, the older group experienced a greater impact than younger adults in all regions. Addressing these disparities requires targeted public health strategies and policies to reduce exposure to PM_{2.5} and mitigate its adverse effects on life expectancy.

CRedit authorship contribution statement

Pattheera Somboonsin: Writing – original draft, Visualization, Methodology, Formal analysis, Data curation, Conceptualization. **Juthasinee Thanyapraneedkul:** Writing – review & editing, Methodology. **Ivan C. Hanigan:** Writing – review & editing, Resources, Methodology. **Vladimir Canudas-Romo:** Writing – review & editing, Methodology, Conceptualization.

Ethical approval

Since this study relied on data from publicly accessible sources and did not involve personal information, it was exempt from requiring ethics approval from our respective Institutional Review Board (IRB).

Funding

The author declare that no funds, grants, or other support were received during the preparation of this manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

We acknowledge that this research was analysed using resources provided by the Clean Air Research Data and Analysis Tools (CARDAT) platform (<https://cardat.github.io>). CARDAT is supported by The Centre

for Safe Air (<https://safeair.org.au/>), with funding from the National Health and Medical Research Council (Grant No. 2015584) and the WHO Collaborating Centre for Climate Change and Health Impact Assessment with funding from Australian Research Data Commons (ARDC) AirHealth Data Bridges project (<https://doi.org/10.47486/PS022>).

We also acknowledge that analyses and map and line chart visualisations of PM_{2.5} used in this study were produced with the Giovanni online data system (<https://giovanni.gsfc.nasa.gov/giovanni/>), developed and maintained by the NASA GES DISC.

We are especially grateful to Professor Fay Johnston from the Menzies Institute for Medical Research, University of Tasmania, for her invaluable comments and suggestions, which have significantly enhanced this manuscript.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.envres.2025.122690>.

Data availability

Data will be made available on request.

References

- AHMD, 2025. Australian human mortality database. In: School of Demography. Australian National University.
- Amini, H., Yousefian, F., Faridi, S., Andersen, Z.J., Calas, E., Castro, A., Cervantes-Martínez, K., Cole-Hunter, T., Corso, M., Dragić, N., 2024. Two decades of air pollution health risk assessment: insights from the use of who's AirQ and AirQ+ tools. *Public Health Rev.* 45, 1606969.
- Andersen, P.K., 2013. Decomposition of number of life years lost according to causes of death. *Stat. Med.* 32 (30), 5278–5285. <https://doi.org/10.1002/sim.5903>.
- Andersen, P.K., Canudas-Romo, V., Keiding, N., 2013. Cause-specific measures of life years lost. *Demogr. Res.* 29, 1127–1152.
- Aragón, E., Watt, S., Nguyen Duc, H., Cheeseman, C., Riley, M., Leys, J., White, S., Salter, D., Azzi, M., Tzu-Chi Chang, L., 2021. Dust transport from inland Australia and its impact on air quality and health on the eastern coast of Australia during the February 2019 dust storm. *Atmosphere* 12 (2), 141.
- Australian Bureau of Statistics, 2021a. Australian Statistical Geography Standard (ASGS) Edition 3. ABS. Retrieved 10 June 2024 from. <https://www.abs.gov.au/statistics/standards/australian-statistical-geography-standard-asgs-edition-3/latest-release>.
- Australian Bureau of Statistics, 2021b. Greater Capital City statistical areas. ABS. Retrieved 2024 June 10 from. <https://www.abs.gov.au/statistics/standards/australian-statistical-geography-standard-asgs-edition-3/jul2021-jun2026/main-structure-and-greater-capital-city-statistical-areas/greater-capital-city-statistical-areas>.
- Australian Bureau of Statistics, 2022. Deaths, Australia. Retrieved 10 June 2024 from. <https://www.abs.gov.au/statistics/people/population/deaths-australia/latest-release>.
- Australian Bureau of Statistics, 2023. Regional population by age and sex. Retrieved 10 June 2024 from. <https://www.abs.gov.au/statistics/people/population/regional-population-age-and-sex/2023>.
- Australian Bureau of Statistics, 2024. National, State and Territory Population. ABS. Retrieved 20 September 2024 from. <https://www.abs.gov.au/statistics/people/population/national-state-and-territory-population/mar-2024>.
- Bai, L., He, Z., Li, C., Chen, Z., 2020. Investigation of yearly indoor/outdoor PM_{2.5} levels in the perspectives of health impacts and air pollution control: case study in Changchun, in the northeast of China. *Sustain. Cities Soc.* 53, 101871. <https://doi.org/10.1016/j.scs.2019.101871>.
- Baillie, C.R., Ghosh, J.K.C., Kirk, M.D., Sullivan, S.G., 2023. Effect of ambient PM_{2.5} on healthcare utilisation for acute respiratory illness, Melbourne, Victoria, Australia, 2014–2019. *J. Air Waste Manag. Assoc.* 73 (2), 120–132. <https://doi.org/10.1080/10962247.2022.2146810>.
- Baklanov, A., Molina, L.T., Gauss, M., 2016. Megacities, air quality and climate. *Atmos. Environ.* 126, 235–249. <https://doi.org/10.1016/j.atmosenv.2015.11.059>.
- Baum, S., Horton, S., Low Choy, D., Gleeson, B., 2009. Climate change, health impacts and urban adaptability: case study of Gold Coast City. *Research Monograph* 11, 68.
- Borchers-Arriagada, N., Morgan, G.G., Van Buskirk, J., Gopi, K., Yuen, C., Johnston, F.H., Guo, Y., Cope, M., Hanigan, I.C., 2024. Daily PM_{2.5} and seasonal-trend decomposition to identify extreme air pollution events from 2001 to 2020 for Continental Australia using a random Forest model. *Atmosphere* 15 (11), 1341. <https://www.mdpi.com/2073-4433/15/11/1341>.
- Bowman, D.M., Borchers-Arriagada, N., Macintosh, A., Butler, D.W., Williamson, G.J., Johnston, F.H., 2024. Climate change must be factored into savanna carbon-management projects to avoid maladaptation: the case of worsening air pollution in western Top End of the Northern Territory, Australia. *The Rangeland Journal* 46, Rj23049. <https://doi.org/10.1071/Rj23049>.
- Bowman, D.M.J.S., Williamson, G.J., Gibson, R.K., Bradstock, R.A., Keenan, R.J., 2021. The severity and extent of the Australia 2019–20 Eucalyptus forest fires are not the legacy of forest management. *Nature Ecology & Evolution* 5 (7), 1003–1010. <https://doi.org/10.1038/s41559-021-01464-6>.
- Broome, R., Powell, P., Cope, M., Morgan, G., 2019. Source-specific impacts of PM_{2.5} on mortality in the Greater Metropolitan Region of Sydney, Australia. *Environmental Epidemiology* 3. https://journals.lww.com/enviroepidem/fulltext/2019/10001/source_specific_impacts_of_pm2_5_on_mortality_in.839.aspx.
- Broome, R.A., Powell, J., Cope, M.E., Morgan, G.G., 2020. The mortality effect of PM_{2.5} sources in the Greater Metropolitan Region of Sydney, Australia. *Environ. Int.* 137, 105429. <https://doi.org/10.1016/j.envint.2019.105429>.
- Buchard, V., Da Silva, A., Randles, C., Colarco, P., Ferrare, R., Hair, J., Hostetler, C., Tackett, J., Winker, D., 2016. Evaluation of the surface PM_{2.5} in Version 1 of the NASA MERRA aerosol Reanalysis over the United States. *Atmos. Environ.* 125, 100–111.
- Bureau of Meteorology, 2019. Explainer: what is a dust storm? Retrieved 01 July 2024 from. <https://media.bom.gov.au/social/blog/1990/explainer-what-is-a-dust-storm/>.
- Butler, C.D., Whelan, J., 2018. Air pollution and climate change in Australia: a triple burden. In: *Climate Change and Air Pollution: the Impact on Human Health in Developed and Developing Countries*, pp. 131–149.
- Canudas-Romo, V., García-Guerrero, V.M., Echarrí-Cánovas, C.J., 2015. The stagnation of the Mexican male life expectancy in the first decade of the 21st century: the impact of homicides and diabetes mellitus. *J. Epidemiol. Community Health* 69 (1), 28–34.
- Cao, K., Tang, X., Wang, L., Chen, X., Wu, Q., Kong, L., Lu, M., Wu, H., Wang, Z., 2023. Simulating the impact of bushfires in Australia on local air quality and aerosol burden in the Southern Hemisphere. *SOLA* 19, 16–25.
- Cao, Y., Wu, J., 2025. Impacts of wildfire smoke PM_{2.5}, greenspace and terrain ruggedness on life expectancy in the contiguous United States. *Sci. Total Environ.* 964, 178584. <https://doi.org/10.1016/j.scitotenv.2025.178584>.
- Chen, C.-C., Wang, Y.-R., Yeh, H.-Y., Lin, T.-H., Huang, C.-S., Wu, C.-F., 2021. Estimating monthly PM_{2.5} concentrations from satellite remote sensing data, meteorological variables, and land use data using ensemble statistical modeling and a random forest approach. *Environ. Pollut.* 291, 118159.
- Chen, J., Hoek, G., 2020. Long-term exposure to PM and all-cause and cause-specific mortality: a systematic review and meta-analysis. *Environ. Int.* 143, 105974. <https://doi.org/10.1016/j.envint.2020.105974>.
- Chen, K., Glonek, G., Hansen, A., Williams, S., Tuke, J., Salter, A., Bi, P., 2016. The effects of air pollution on asthma hospital admissions in Adelaide, South Australia, 2003–2013: time-series and case-crossover analyses. *Clin. Exp. Allergy* 46 (11), 1416–1430.
- Chen, L., Wang, L., Qian, Y., Chen, H., 2022. Changes and trend disparities in life expectancy and health-adjusted life expectancy attributed to disability and mortality from 1990 to 2019 in China. *Front. Public Health* 10. <https://doi.org/10.3389/fpubh.2022.925114>. Original Research.
- Cheney, N.P., Gould, J.S., McCaw, W.L., Anderson, W.R., 2012. Predicting fire behaviour in dry eucalypt forest in southern Australia. *For. Ecol. Manag.* 280, 120–131. <https://doi.org/10.1016/j.foreco.2012.06.012>.
- Chidubee, P., Nusi, K., Tantane, S., 2023. Estimation of PM_{2.5} surface concentrations in the North-eastern and Eastern regions of Thailand from NASA's MERRA2 satellite data. *IOP Conf. Ser. Earth Environ. Sci.*
- Chu, Y., Liu, Y., Li, X., Liu, Z., Lu, H., Lu, Y., Mao, Z., Chen, X., Li, N., Ren, M., 2016. A review on predicting ground PM_{2.5} concentration using satellite aerosol optical depth. *Atmosphere* 7 (10), 129.
- Collins, L., Bradstock, R.A., Clarke, H., Clarke, M.F., Nolan, R.H., Penman, T.D., 2021. The 2019/2020 mega-fires exposed Australian ecosystems to an unprecedented extent of high-severity fire. *Environ. Res. Lett.* 16 (4), 044029. <https://doi.org/10.1088/1748-9326/abeb9e>.
- Country Fire Authority, 2024. Major fires. <https://www.cfa.vic.gov.au/about-us/history-major-fires/major-fires>.
- Cunningham, S., McCutcheon, M., Hearn, G., Ryan, M.D., Collis, C., 2019. Australian Cultural and Creative Activity: a Population and Hotspot Analysis. Gold Coast.
- de Jesus, A.L., Thompson, H., Knibbs, L.D., Hanigan, I., De Torres, L., Fisher, G., Berko, H., Morawska, L., 2020. Two decades of trends in urban particulate matter concentrations across Australia. *Environ. Res.* 190, 110021.
- De Marco, A., Amoatey, P., Khaniabadi, Y.O., Sicard, P., Hopke, P.K., 2018. Mortality and morbidity for cardiopulmonary diseases attributed to PM_{2.5} exposure in the metropolis of Rome, Italy. *Eur. J. Intern. Med.* 57, 49–57. <https://doi.org/10.1016/j.ejim.2018.07.027>.
- Dirgawati, M., Hinwood, A., Nedkoff, L., Hankey, G.J., Yeap, B.B., Flicker, L., Nieuwenhuijsen, M., Brunekreef, B., Heyworth, J., 2019. Long-term exposure to low air pollutant concentrations and the relationship with all-cause mortality and stroke in older men. *Epidemiology* 30, S82–S89. <https://doi.org/10.1097/ede.0000000000001034>.
- Egerstrom, N., Rojas-Rueda, D., Martuzzi, M., Jalaludin, B., Nieuwenhuijsen, M., So, R., Lim, Y.H., Loft, S., Andersen, Z.J., Cole-Hunter, T., 2023. Health and economic benefits of meeting WHO air quality guidelines, Western Pacific Region. *Bull. World Health Organ.* 101 (2), 130–139. <https://doi.org/10.2471/BLT.22.288938>.
- Erlangsen, A., Andersen, P.K., Toender, A., Laursen, T.M., Nordentoft, M., Canudas-Romo, V., 2017. Cause-specific life-years lost in people with mental disorders: a nationwide, register-based cohort study. *Lancet Psychiatry* 4 (12), 937–945.
- Fielke, S.J., Bardsley, D.K., 2015. Regional agricultural governance in peri-urban and rural South Australia: strategies to improve multifunctionality. *Sustain. Sci.* 10 (2), 231–243. <https://doi.org/10.1007/s11625-014-0272-6>.

- Forest Fire Management Victoria, 2023. Past bushfires: a chronology of major bushfires in Victoria from 2020 back to 1851. Retrieved 01 July 2024 from. <https://www.ffm.vic.gov.au/history-and-incidents/past-bushfires>.
- Godfree, R.C., Knerr, N., Encinas-Viso, F., Albrecht, D., Bush, D., Christine Cargill, D., Clements, M., Gueidan, C., Guja, L.K., Harwood, T., Joseph, L., Lepschi, B., Nargar, K., Schmidt-Lebuhn, A., Broadhurst, L.M., 2021. Implications of the 2019–2020 megafires for the biogeography and conservation of Australian vegetation. *Nat. Commun.* 12 (1), 1023. <https://doi.org/10.1038/s41467-021-21266-5>.
- Golam Mortoja, M., Yigitcanlar, T., 2021. Public perceptions of peri-urbanism triggered climate change: survey evidence from South East Queensland, Australia. *Sustain. Cities Soc.* 75, 103407. <https://doi.org/10.1016/j.scs.2021.103407>.
- Government of the Commonwealth of Australia - Australian Bureau of, S., 2023. ABS - life tables - life expectancy (GCCSA) 2010–2019 Australian Urban Research Infrastructure Network (AURIN). <https://data.aurin.org.au/dataset/542ad516-8d4b-46ae-8cef-47ab3afa6819>. <https://researchdata.edu.au/abs-life-tables-2010-2019>.
- Graham, A.M., Pringle, K.J., Pope, R.J., Arnold, S.R., Conibear, L.A., Burns, H., Rigby, R., Borchers-Arriagada, N., Butt, E.W., Kiely, L., Reddington, C., Spracklen, D.V., Woodhouse, M.T., Knote, C., McQuaid, J.B., 2021. Impact of the 2019/2020 Australian megafires on air quality and health. *GeoHealth* 5 (10), e2021GH000454. <https://doi.org/10.1029/2021GH000454>.
- Hadei, M., Shamsavani, A., Krzyzanowski, M., Querol, X., Stafoggia, M., Nazari, S.S.H., Jafari, A.J., Yarahmadi, M., Kermani, M., Khosravi, A., 2020. Burden of mortality attributed to PM2.5 exposure in cities of Iran; contribution of short-term pollution peaks. *Atmos. Environ.* 224, 117365. <https://doi.org/10.1016/j.atmosenv.2020.117365>.
- Hanigan, I., Yuen, C., Gopi, K., Borchers-Arriagada, N., van Buskirk, J., Morgan, G., 2023. Bushfire specific PM2.5 Aus 2001–2020 v1.3. Centre for Air pollution, energy and health Research. <https://doi.org/10.17605/OSF.IO/WQK4T>.
- Hanigan, I.C., Broome, R.A., Chaston, T.B., Cope, M., Dennekamp, M., Heyworth, J.S., Heathcote, K., Horsley, J.A., Jalaludin, B., Jegasothy, E., 2021. Avoidable mortality attributable to anthropogenic fine particulate matter (PM2.5) in Australia. *Int. J. Environ. Res. Publ. Health* 18 (1), 254.
- Hanigan, I.C., Morgan, G.G., Williamson, G.J., Salimi, F., Henderson, S.B., Turner, M.R., Bowman, D.M., Johnston, F.H., 2018. Extensible database of validated biomass smoke events for health research. *Fire* 1 (3), 50. <https://doi.org/10.3390/fire1030050>.
- Hanigan, I.C., Rolf, M.L., Knibbs, L.D., Salimi, F., Cowie, C.T., Heyworth, J., Marks, G.B., Guo, Y., Cope, M., Bauman, A., Jalaludin, B., Morgan, G.G., 2019. All-cause mortality and long-term exposure to low level air pollution in the '45 and up study' cohort, Sydney, Australia, 2006–2015. *Environ. Int.* 126, 762–770. <https://doi.org/10.1016/j.envint.2019.02.044>.
- Hansen, A., Bi, P., Nitschke, M., Pisanelli, D., Ryan, P., Sullivan, T., Barnett, A.G., 2012. Particulate air pollution and cardiorespiratory hospital admissions in a temperate Australian city: a case-crossover analysis. *Sci. Total Environ.* 416, 48–52. <https://doi.org/10.1016/j.scitotenv.2011.09.027>.
- Haswell, M., Bethmont, A., 2016. Health concerns associated with unconventional gas mining in rural Australia. *Rural Remote Health* 16 (4), 1–9. <https://search.informit.com/doi/10.3316/informit.206863265534988>.
- Healy, E., 2012. Is an 'exponential decline' in car use in Melbourne a plausible greenhouse abatement measure? *Aust. Plan.* 49 (1), 88–98. <https://doi.org/10.1080/07293682.2011.608690>.
- Hefferan, M., Mannix, J., 2011. Sunshine Coast Demographics: More than Just the Numbers.
- Hertzog, L., Morgan, G.G., Yuen, C., Gopi, K., Pereira, G.F., Johnston, F.H., Cope, M., Chaston, T.B., Vyas, A., Vardoulakis, S., 2024. Mortality burden attributable to exceptional PM2.5 air pollution events in Australian cities: a health impact assessment. *Heliyon* 10 (2), e24532. <https://doi.org/10.1016/j.heliyon.2024.e24532>.
- Hill, S., Cumpston, Z., Vigiola, G., 2021. Australia State of the Environment 2021: Urban Independent Report to the Australian Government Minister for the Environment. Commonwealth of Australia, Canberra. DOI, 10.
- HMD, 2025. Human Mortality Database. Max Planck Institute for Demographic Research (Germany). University of California, Berkeley (USA) and French Institute for Demographic Studies (France). www.mortality.org.
- Houle, B., Kabudula, C., Mojola, S.A., Angotti, N., Gómez-Olivé, F.X., Gareta, D., Herbst, K., Clark, S.J., Menken, J., Canudas-Romo, V., 2024. Mortality variability and differentials by age and causes of death in rural South Africa, 1994–2018. *BMJ Glob. Health* 9 (4), e013539. <https://doi.org/10.1136/bmjgh-2023-013539>.
- IQAir, 2023. 2023 world air quality report: region and City PM2.5 ranking. <https://www.iqair.com/dl/2023-World-Air-Quality-Report.pdf>.
- Jegasothy, E., Hanigan, I.C., Van Buskirk, J., Morgan, G.G., Jalaludin, B., Johnston, F.H., Guo, Y., Broome, R.A., 2023. Acute health effects of bushfire smoke on mortality in Sydney, Australia. *Environ. Int.* 171, 107684. <https://doi.org/10.1016/j.envint.2022.107684>.
- Johnson, A.L., Gao, C.X., Dennekamp, M., Williamson, G.J., Brown, D., Carroll, M.T.C., Ikin, J.F., Del Monaco, A., Abramson, M.J., Guo, Y., 2019. Associations between respiratory health outcomes and coal mine fire PM2.5 smoke exposure: a cross-sectional study. *Int. J. Environ. Res. Publ. Health* 16 (21), 4262. <https://doi.org/10.3390/ijerph16214262>.
- Johnston, F.H., Borchers-Arriagada, N., Morgan, G.G., Jalaludin, B., Palmer, A.J., Williamson, G.J., Bowman, D.M.J.S., 2021. Unprecedented health costs of smoke-related PM2.5 from the 2019–20 Australian megafires. *Nat. Sustain.* 4 (1), 42–47. <https://doi.org/10.1038/s41893-020-00610-5>.
- Johnston, F.H., Hanigan, I.C., Henderson, S.B., Morgan, G.G., 2013. Evaluation of interventions to reduce air pollution from biomass smoke on mortality in Launceston, Australia: retrospective analysis of daily mortality, 1994–2007. *BMJ Br. Med. J. (Clin. Res. Ed.)* 346, e8446. <https://doi.org/10.1136/bmj.e8446>.
- Knibbs, L.D., van Donkelaar, A., Martin, R.V., Bechle, M.J., Brauer, M., Cohen, D.D., Cowie, C.T., Dirgawati, M., Guo, Y., Hanigan, I.C., Johnston, F.H., Marks, G.B., Marshall, J.D., Pereira, G., Jalaludin, B., Heyworth, J.S., Morgan, G.G., Barnett, A.G., 2018. Satellite-Based land-use regression for continental-scale long-term ambient PM2.5 exposure assessment in Australia. *Environ. Sci. Technol.* 52 (21), 12445–12455. <https://doi.org/10.1021/acs.est.8b02328>.
- Lelieveld, J., Pozzer, A., Pöschl, U., Fnais, M., Haines, A., Münzel, T., 2020. Loss of life expectancy from air pollution compared to other risk factors: a worldwide perspective. *Cardiovasc. Res.* <https://doi.org/10.1093/cvr/cvaa073>.
- Li, C., Martin, R.V., van Donkelaar, A., Boys, B.L., Hammer, M.S., Xu, J.-W., Marais, E.A., Reff, A., Strum, M., Ridley, D.A., 2017. Trends in chemical composition of global and regional population-weighted fine particulate matter estimated for 25 years. *Environ. Sci. Technol.* 51 (19), 11185–11195.
- Li, Y., Yuan, S., Fan, S., Song, Y., Wang, Z., Yu, Q., Yu, Q., Liu, Y., 2021. Satellite remote sensing for estimating PM 2.5 and its components. *Curr. Pollut. Rep.* 7, 72–87.
- Lim, C.-H., Ryu, J., Choi, Y., Jeon, S.W., Lee, W.-K., 2020. Understanding global PM2.5 concentrations and their drivers in recent decades (1998–2016). *Environ. Int.* 144, 106011. <https://doi.org/10.1016/j.envint.2020.106011>.
- Liu, T., Meng, H., Yu, M., Xiao, Y., Huang, B., Lin, L., Zhang, H., Hu, R., Hou, Z., Xu, Y., 2021a. Urban-rural disparity of the short-term association of PM2.5 with mortality and its attributable burden. *Innovation* 2 (4).
- Liu, X., Mubarik, S., Wang, F., Yu, Y., Wang, Y., Shi, F., Wen, H., Yu, C., 2021b. Lung cancer death attributable to long-term ambient Particulate Matter (PM2.5) exposure in East Asian countries during 1990–2019. *Front. Med.* 8. <https://doi.org/10.3389/fmed.2021.742076>.
- NASA, 2006. Dust storm in the Simpson Desert, Australia. Retrieved 01 July 2024 from. <https://earthobservatory.nasa.gov/images/17417/dust-storm-in-the-simpson-desert-australia>.
- NASA, 2009. Dust over Eastern Australia. Retrieved 01 July 2024 from. <https://earthobservatory.nasa.gov/images/event/40275>.
- NSW Office of Environment and Heritage, 2022. DustWatch data explorer. Retrieved 01 July 2024 from. <https://www.environment.nsw.gov.au/dustwatchapp/default.aspx>.
- NSW rural fire service, 2014. Australian Bush fire history. Retrieved 01 July 2024 from. <https://canyonleigh.rfsa.org.au/2014/08/12/australian-bush-fire-history/>.
- Ntourou, K., Fameli, K.-M., Moustiris, K., Augoustinos, A., Tsisits, C., 2023a. The influence of ozone concentrations on public health over the greater Athens area, Greece. *Environmental Sciences Proceedings* 26 (1), 107. <https://www.mdpi.com/2673-4931/26/1/107>.
- Ntourou, K., Moustiris, K., Spyropoulos, G., Fameli, K.-M., Manousakis, N., 2023b. Adverse health effects (Bronchitis cases) due to particulate matter exposure: a twenty-year scenario analysis for the greater Athens area (Greece) using the AirQ+ model. *Atmosphere* 14 (2), 389. <https://www.mdpi.com/2073-4433/14/2/389>.
- Palmer, D., Koubli, E., Cole, I., Betts, T., Gottschalg, R., 2018. Satellite or ground-based measurements for production of site specific hourly irradiance data: which is most accurate and where? *Sol. Energy* 165, 240–255.
- Parslow, R., Jorm, A., Christensen, H., Jacomb, P., Rodgers, B., 2004. Gender differences in factors affecting use of health services: an analysis of a community study of middle-aged and older Australians. *Soc. Sci. Med.* 59 (10), 2121–2129. <https://doi.org/10.1016/j.socscimed.2004.03.018>.
- Pinichka, C., Makka, N., Sukkumnoed, D., Chariyalertsak, S., Inchai, P., Bundhamcharoen, K., 2017. Burden of disease attributed to ambient air pollution in Thailand: a GIS-based approach. *PLoS One* 12 (12), e0189909.
- Preston, S., Heuveline, P., Guillot, M., 2000. *Demography: Measuring and Modeling Population Processes*. Blackwell Publishers, Malden, MA, pp. 71–91.
- Regan, J.C., Partridge, L., 2013. Gender and longevity: why do men die earlier than women? Comparative and experimental evidence. *Best Pract. Res. Clin. Endocrinol. Metabol.* 27 (4), 467–479. <https://doi.org/10.1016/j.beem.2013.05.016>.
- Romsey Australia. Historical Bushfires in Australia. Retrieved 01 July 2024 from. <http://romseyaustralia.com/fireall.html>.
- Rovira, J., Domingo, J.L., Schuhmacher, M., 2020. Air quality, health impacts and burden of disease due to air pollution (PM10, PM2.5, NO2 and O3): application of AirQ+ model to the Camp de Tarragona County (Catalonia, Spain). *Sci. Total Environ.* 703, 135538. <https://doi.org/10.1016/j.scitotenv.2019.135538>.
- Ryan, R.G., Silver, J.D., Schofield, R., 2021. Air quality and health impact of 2019–20 Black Summer megafires and COVID-19 lockdown in Melbourne and Sydney, Australia. *Environ. Pollut.* 274, 116498. <https://doi.org/10.1016/j.envpol.2021.116498>.
- Sicard, P., Khaniabadi, Y.O., Perez, S., Gualtieri, M., De Marco, A., 2019. Effect of O3, PM10 and PM2.5 on cardiovascular and respiratory diseases in cities of France, Iran and Italy. *Environ. Sci. Pollut. Control Ser.* 26 (31), 32645–32665. <https://doi.org/10.1007/s11356-019-06445-8>.
- Somboonsin, P., Houle, B., Canudas-Romo, V., 2024. Sex disparities in the burden of air particulate matter-related unhealthy years and life-years lost in Asia-Pacific countries, 1990–2019. *Atmos. Environ.*, 120763. <https://doi.org/10.1016/j.atmosenv.2024.120763>.
- Somboonsin, P., Vardoulakis, S., Canudas-Romo, V., 2023. A comparative study of life-years lost attributable to air particulate matter in Asia-Pacific and European countries. *Chemosphere* 338, 139420. <https://doi.org/10.1016/j.chemosphere.2023.139420>.
- The South Australian Country Fire Service, 2023. Bushfire history. Retrieved 01 July 2024 from. <https://www.cfs.sa.gov.au/about/about/bushfire-history/>.

- United Nations, 2015. Transforming our world: the 2030 agenda for sustainable development. Retrieved 04 September 2024 from. <https://sdgs.un.org/publications/transforming-our-world-2030-agenda-sustainable-development-17981>.
- United Nations, 2024. Synergy solutions for climate and SDG action: bridging the ambition gap for the future we want. Retrieved September, 04 from. <https://sdgs.un.org/publications/synergy-solutions-climate-and-sdg-action-bridging-ambition-gap-future-we-want-56106>.
- Walter, C.M., Schneider-Futschik, E.K., Lansbury, N.L., Sly, P.D., Head, B.W., Knibbs, L.D., 2021. The health impacts of ambient air pollution in Australia: a systematic literature review. *Intern. Med. J.* 51 (10), 1567–1579.
- Williamson, K., Satre-Meloy, A., Velasco, K., Green, K., 2018. Climate Change Needs Behavior Change: Making the Case for Behavioral Solutions to Reduce Global Warming. Rare, Arlington, VA, USA.
- World Health Organization, 2020a. Airq+: Burden of Disease due to Air Pollution Manual.
- World Health Organization, 2020b. Airq+: Health Impact Assessment of Air Pollution: Airq+ multiple-area Data Input.
- Yu, W., Guo, Y., Shi, L., Li, S., 2020. The association between long-term exposure to low-level PM_{2.5} and mortality in the state of Queensland, Australia: a modelling study with the difference-in-differences approach. *PLoS Med.* 17 (6), e1003141.
- Zeydan, Ö., Wang, Y., 2019. Using MODIS derived aerosol optical depth to estimate ground-level PM_{2.5} concentrations over Turkey. *Atmos. Pollut. Res.* 10 (5), 1565–1576.
- Zhao, Q., Yu, P., Mahendran, R., Huang, W., Gao, Y., Yang, Z., Ye, T., Wen, B., Wu, Y., Li, S., Guo, Y., 2022. Global climate change and human health: pathways and possible solutions. *Eco-Environ. Health* 1 (2), 53–62. <https://doi.org/10.1016/j.eehl.2022.04.004>.